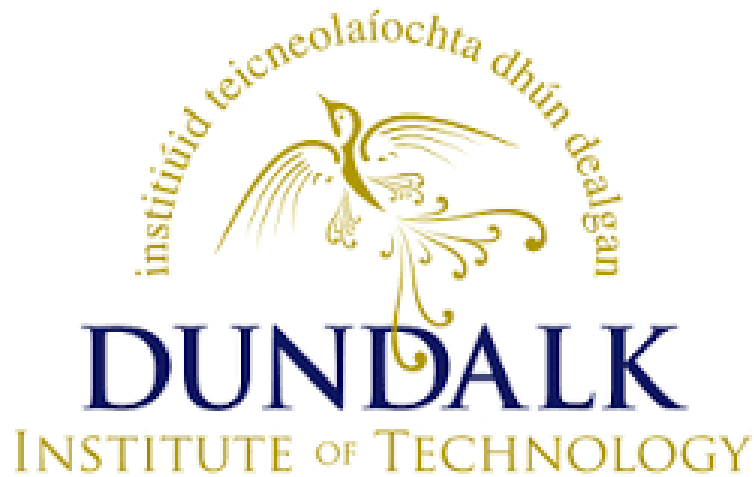




Time Series Analysis

Continuous Assessment 2

Final Report

**Title: ARIMA, SARIMA and Feedforward Neural Network Forecasting
Of Brent Crude Oil Prices and MYR/USD Exchange Rate:
A Perspective of Malaysian Market (2019–2026)**

COMPC8136

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Project Overview

This project analyses 2 financially interconnected time series data given that

1. Brent Crude Oil is the global oil benchmark which directly impacting Malaysia's GDP, PETRONAS revenue and its monetary policy decision of Bank Negara Malaysia which is the central bank of Malaysia
2. MYR/USD Exchange rate is Malaysia most strategically monitored currency pair managed by Bank Negara Malaysia

This report will only include the summarization of the full pipeline from the data acquisition, EDA, decomposition, stationary testing, ARIMA/SARIMA modelling and forecasting as well as the regularized Feedforward Neural Network with dropout and early stopping for 30 days ahead forecasting. For the full interpretation you may refer to the jupyter notebook named "CA2_tsa_PeiWenTanWendy_d00253240.ipynb".

Step 1: Data Description and Acquisition

Data Source and Overview

Both datasets are sourced from Yahoo Finance through the yfiance python library. The study focuses on 2 financially significant Malaysia market time series which are the Brent Crude Oil with the ticker of BZ=F and the MYR/USD Exchange Rate with the ticker of MYR=X over a 7 year window from 21st of March 2019 to 20th March 2026. Both of the time series datasets consist of total of 1827 business day observations.

Table of datasets structure for Brent Crude Oil with Ticker of BZ=F

Columns	Description	Frequency	Data Type	Observations
Date	Format: YYYY-MM-DD	Business Day	Date	1827
Brent_USD	Closing Price in USD per barrel	Business Day	Decimal or Floats	1827

Table of datasets structure for MYR/USD Exchange Rate with the ticker of MYR=X

Columns	Description	Frequency	Data Type	Observations
Date	Format: YYYY-MM-DD	Business Day	Date	1827
USDMYR	Raw value represents MYR per 1 USD	Business Day	Decimal or Floats	1827

Data Cleaning

Firstly, **Missing value is identified and address according** given that the Yahoo Finance return null for non-trading days and public holidays. Each time series data was first review to ensure that there is no null observations through the `.dropna()` then the data is reindexed to business day 'B' frequency using the `.asfreq('B')`. The remaining gaps are filled by forward filling to carry the last observed price forward. **After this preprocessing step both of the time series data has no missing value with total of 1827 observation.**

Secondly, **date formatting** was performed given that the Date columns is automatically handle by the `yfinance` and converted into a standard datetime format without any time zones listed therefore no extra formatting steps were required. To keep the data consistent it was then reindexed to include only the business days to make sure that there are no gaps in the timeline.

Finally, **outlier detection** was carried out to identify any possible outliers contained in both of the time series data. The outliers were identified using the Interquartile Range rule by observe if there were any data points lying beyond the $1.5 \times$ of the IQR below 1st Quartile or above the 3rd Quartile. For Brent Crude Oils datasets there are total of 87 outliers were identified while there were no outliers found for MYR/USD Exchange Rate. My decision was no observation from the outliers are removed as these value represents genuine market conditions rather than data errors and also there were only 87 outliers which composes around 4.7% of the datasets therefore there is no necessary to remove those data points

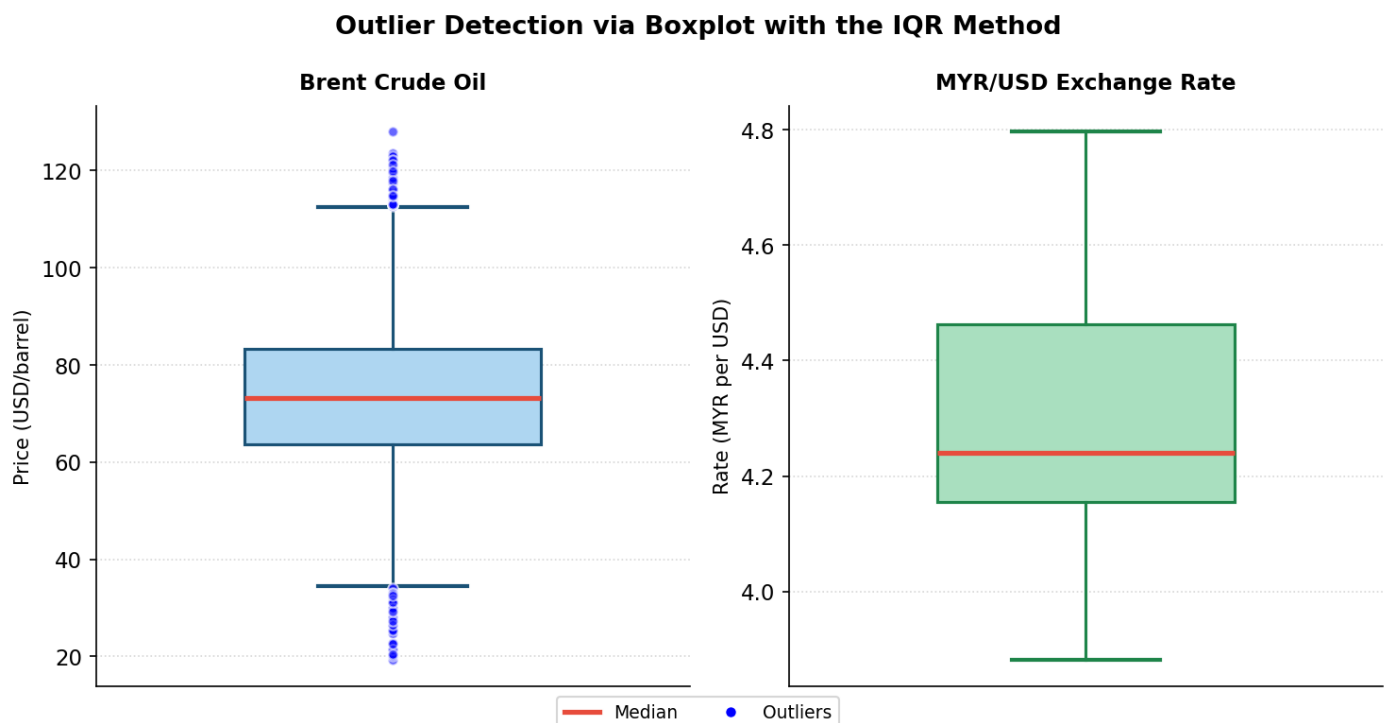


Figure 1: Boxplot of Outlier Detection (IQR Method)

Descriptive Statistics

Statistics	Brent Crude Oil (USD per Barrel)	MYR/USD Exchange Rate
Mean	72.82	4.3231
Standard Deviation	17.84	0.2184
Min	19.33	3.8820
Quartile 1	63.70	4.1553
Median	73.19	4.2400
Quartile 3	83.26	4.4625
Max	127.98	4.7963

Step 2: Exploratory Data Analysis (EDA)

2.1 Visualization of Raw Time Series

The raw daily time series for Brent Crude Oil prices (USD/barrel) and the MYR/USD exchange rate were examined over the full study period from 21 March 2019 to 20 March 2026 in 7 years. Both series show a clear non-stationarity time series which justify the need for differencing prior to ARIMA modelling.

For Brent Crude Oil time series has shown an approximately \$67 per barrel in 2019 and remained relatively stable until early 2020. The COVID-19 pandemic shock happened in March 2020 to May 2020 has caused a steep decline to \$19.33 per barrel driven by collapsing global demand and an oil price war. Prices recovered through year of 2020 to 2021 followed by the geopolitical war spike likely occurred during Feb 2022 to Jun 2022 which pushed prices to a highest price of \$127.98 per barrel. The series stabilized around range of \$70 to \$90 per barrel from 2023 to 2025 with a sharp geopolitical-driven surge in early 2026 exceeding \$110 per barrel.

For MYR/USD Exchange Rate time series, the rate opened at around RM4.05 per USD in March 2019 and briefly weakened to around RM4.35 per USD during the COVID-19 pandemic shock. A sustained depreciation occurred during the US Federal Reserve rate hike cycle occurred in Feb 2022 to Jan 2023 reaching a peak of around RM4.75 per USD. From 2024 onward, the MYR recovered and reaching approximately RM3.88 per USD by early 2026 which is the lowest level in this 7 years study period.

**Figure 1: Raw Time Series of Brent Crude Oil and MYR/USD Exchange Rate
From 21 Mar 2019 to 20 Mar 2026**

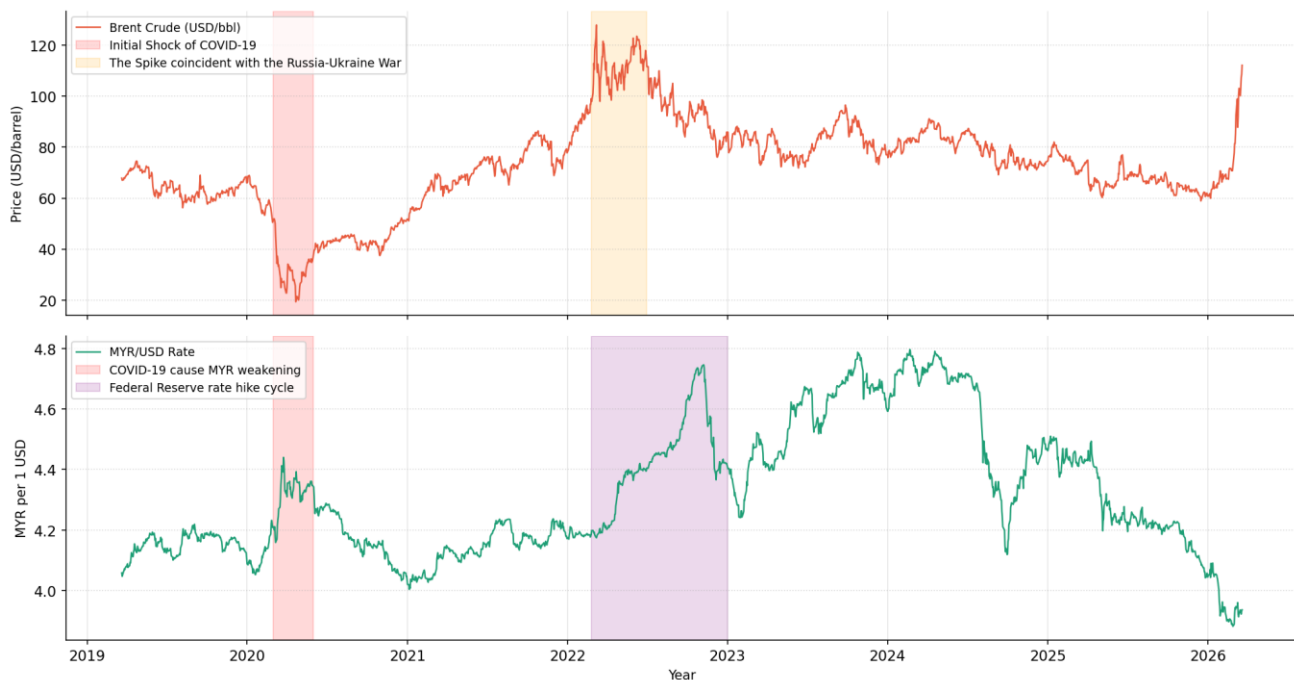


Figure 2: Raw Time Series for Brent Crude Oil and MYR/USD from 21 Mar 2019 to 20 Mar 2026

2.2 Rolling Statistics of Mean and Standard Deviations

A 30 days roiling mean and standard deviation are computed to show trends and volatility changes over time.

Brent crude oil experienced a sharp decline in early 2020 dropping from around \$70 to \$25 per barrel due to the COVID-19 pandemic and an oil price war. Prices then recovered through 2021 peaking near \$115 in middle of 2022 before stabilizing between \$70 and \$85 from late 2022. A new spike appears in early 2026. Volatility was highest during March 2020 to April 2020 and middle 2022 indicating significant price instability in those periods.

**Figure 3a: Rolling Statistics of Brent Crude Oil (USD/barrel)
From 21 Mar 2019 to 20 Mar 2026**

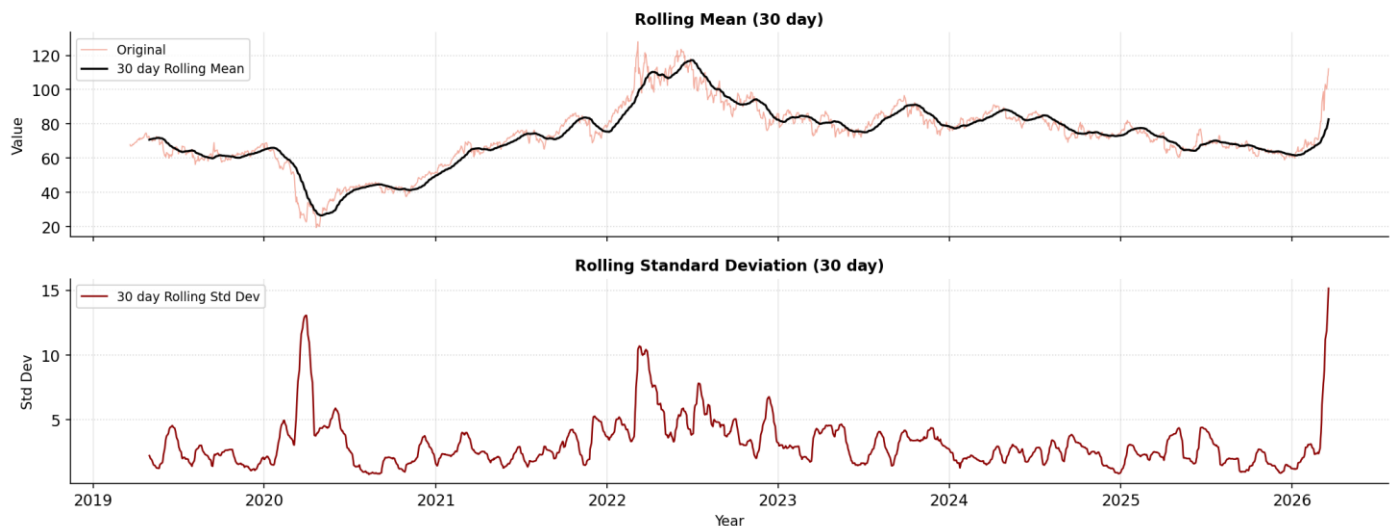


Figure 3a: Rolling Statistics for Brent Crude Oil

The MYR/USD exchange rate shows a gradual decrease of the Malaysian Ringgit from around RM4.05 per USD in year 2019 to a spike near RM4.35 per USD in early 2020 during COVID-19 disruptions. It continued rising to a peak of about RM4.75 per USD in 2022 to 2023, influenced by US interest rate hikes before recovering to around RM3.90 per USD by early 2026. The volatility was moderate with the highest fluctuations occurring in early 2020 and late 2022 to late 2024.

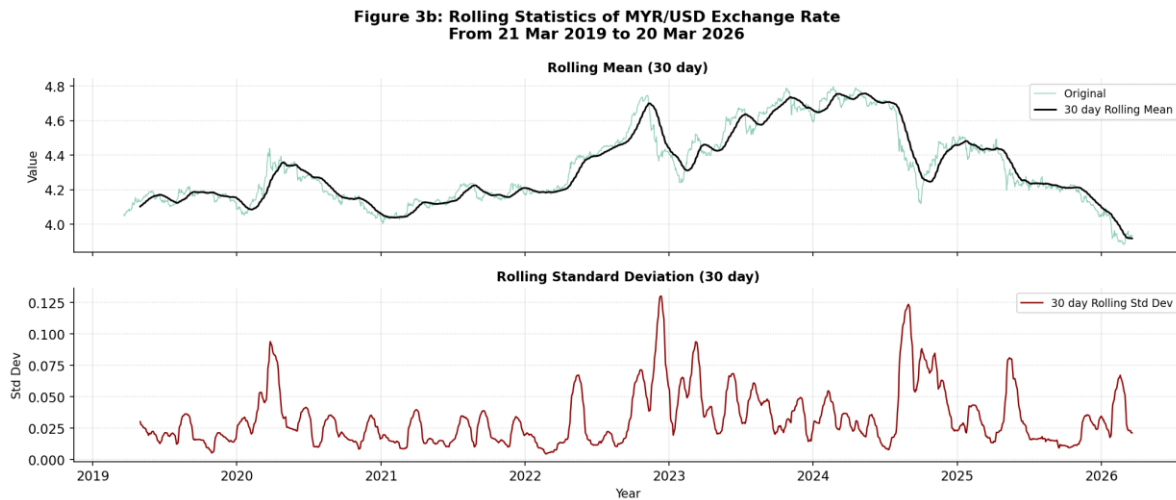


Figure 3b: Rolling Statistics for MYR/USD Exchange Rate

2.3 Classical and STL Decomposition

Both the classical and the STL decomposition is applied to separate the trend, seasonality and residual components for both of the time series data.

2.3.a Classical Decomposition

Classical additive decomposition applied to both of the time series at weekly frequency with period = 52 to separates both series into trend, seasonality and residuals.

For **Brent crude oil**, the trend declines from about \$65 in 2019 to around \$43 in middle of 2020 due to COVID-19 then recovers to a peak near \$97 in 2022 before decreasing to roughly \$68 by early 2026. Seasonality is weak and fluctuating within $\pm \$4$. Residuals show major shocks with a sharp negative spike in early 2020 and a strong positive spike in middle of 2022.

Figure: Classical Additive Decomposition from Brent Crude Oil (USD/barrel)
From 21 Mar 2019 to 20 Mar 2026

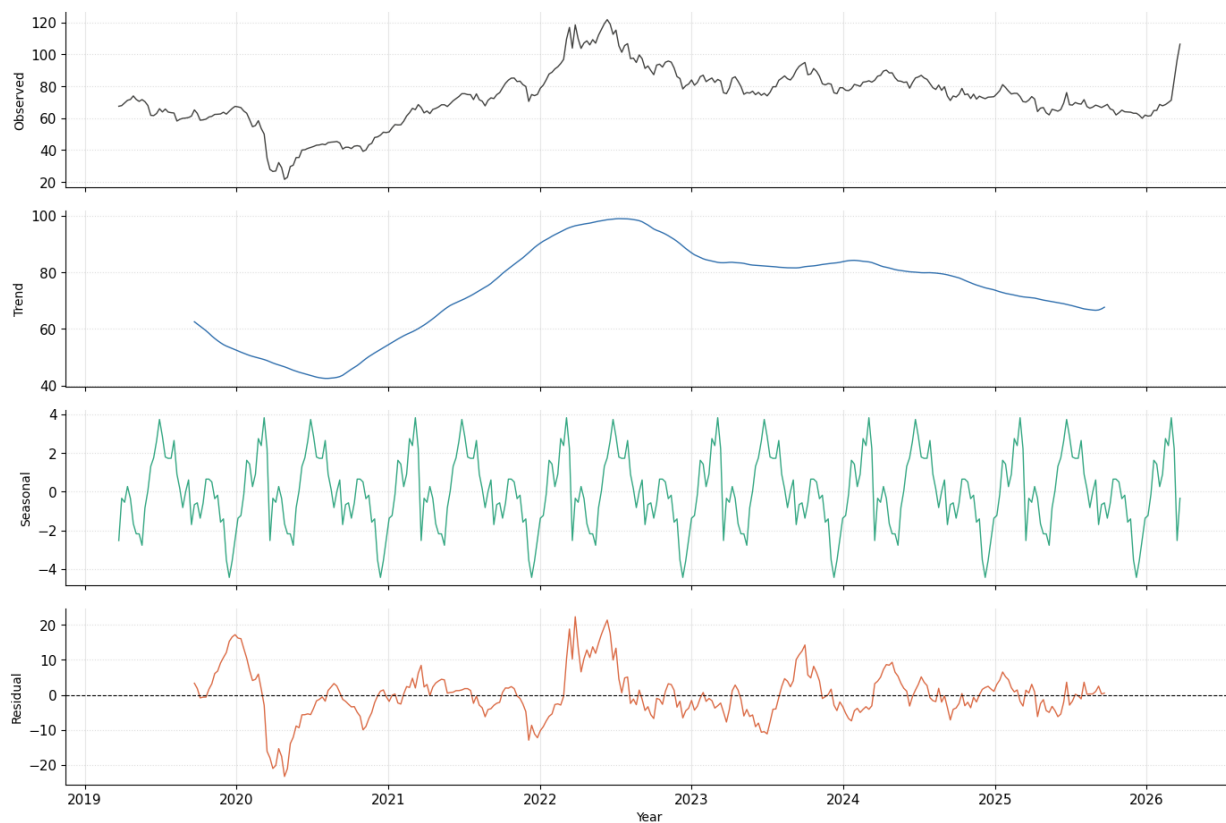


Figure 4a: Classical Decomposition for Brent Crude Oil

For **MYR/USD**, the trend shows gradual depreciation from around RM4.15 per USD in 2019 rising to a peak near RM4.68 per USD in late 2023 due to US rate hikes before recovering to about RM4.20 per USD by early of 2026. Seasonality is mild and ranging roughly between -0.075 and $+0.050$. Residuals are generally small with notable negative deviations in early 2020 and middle of 2024 indicating unexpected short-term strengthening of the MYR.

**Figure: Classical Additive Decomposition from MYR/USD Exchange Rate
From 21 Mar 2019 to 20 Mar 2026**

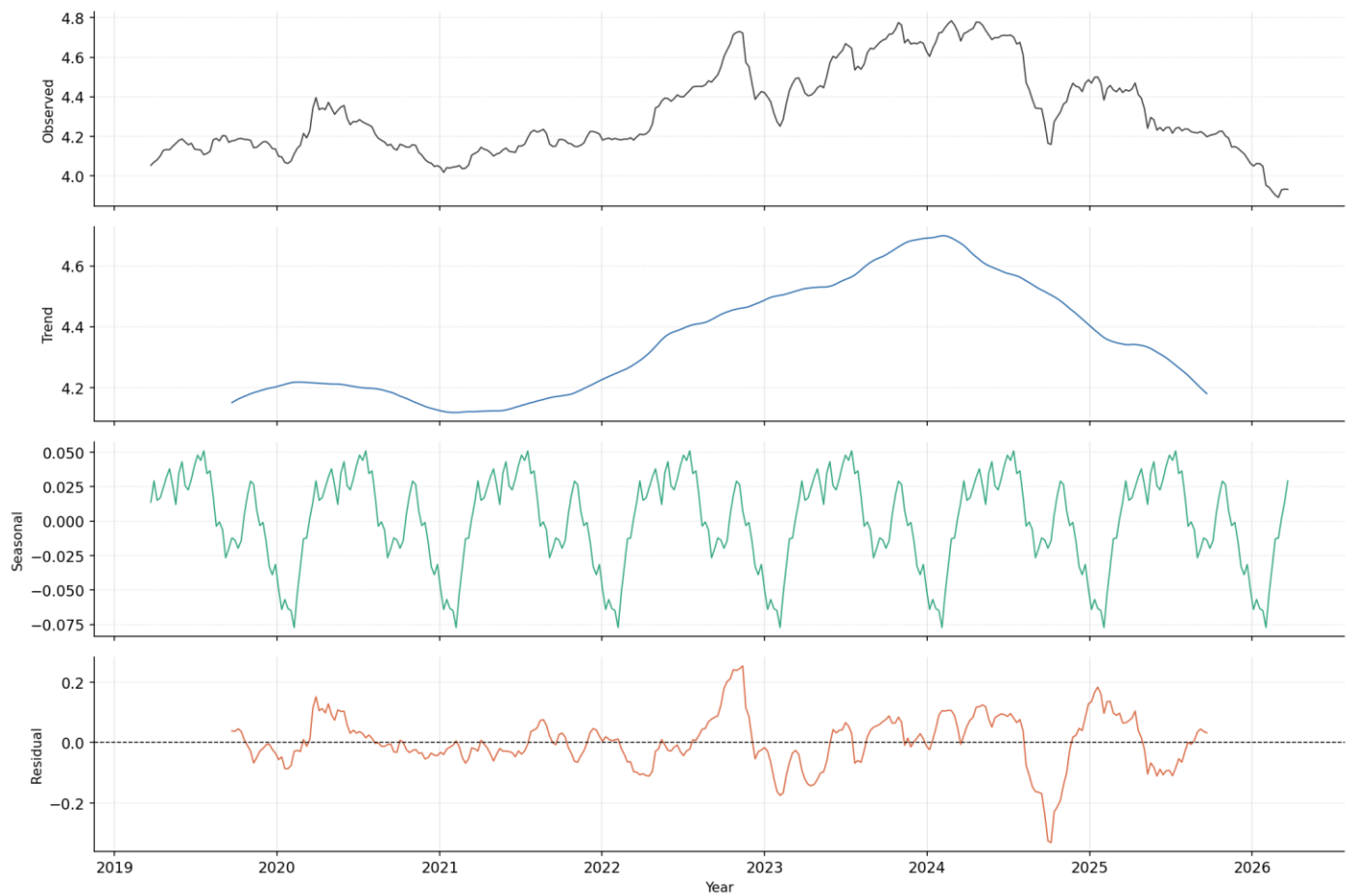


Figure 4b: Classical Decomposition for MYR/USD Exchange Rate

2.3.b STL Decomposition

Seasonal and Trend decomposition using Loess (STL) is carried out for this. The STL decomposition with robust=True is applied to both of the time series data at a weekly frequency with period = 52 to separate the observed data into trend, seasonality and residual components. Unlike the Classical the STL using a fitting procedure that reduce the influence of extreme outliers on the estimate of trend and seasonal.

In this subsections, I will show the key differences between the Classical and STL decomposition

Aspects	Classical Decomposition	STL Decomposition
Trend Estimation	Has centered moving average which leaves null value at both ends of the series	Loess smoothing is applied therefore produced a complete trend across the full series
Seasonal Components	Fixed and Identical each year	Allow to evolve slowly over time
Handling Outliers	Sensitive with extreme value that could possibly distort trend and seasonal estimates	Robust using the 'robust=True' to downweigh outliers during the fitting
Residual	Capture less of spike due to the outliers leakage to the trend	Isolate the spike more cleanly into the residual components
Flexibility	Less flexible but easy to interpret	More flexible but better suitable for long series with more fluctuates pattern

Figure: STL Decomposition Brent Crude Oil (USD/barrel)
From 21 Mar 2019 to 20 Mar 2026

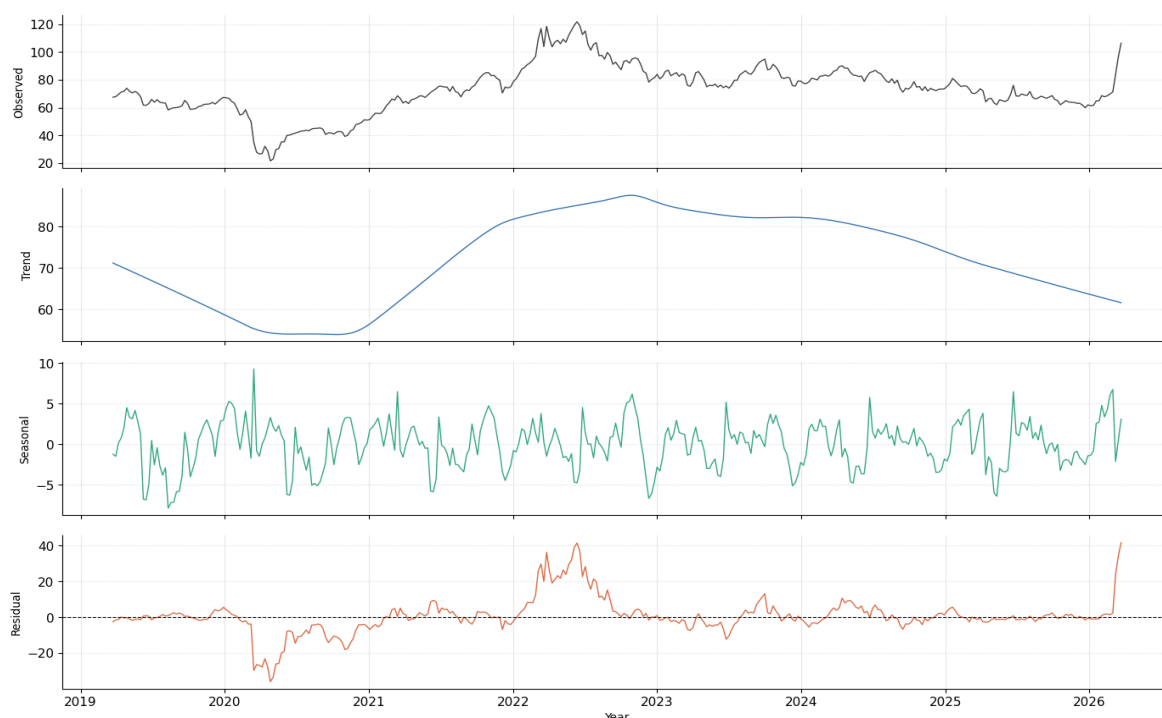


Figure 5a: STL Decomposition for Brent Crude Oil

**Figure: STL Decomposition MYR/USD Exchange Rate
From 21 Mar 2019 to 20 Mar 2026**

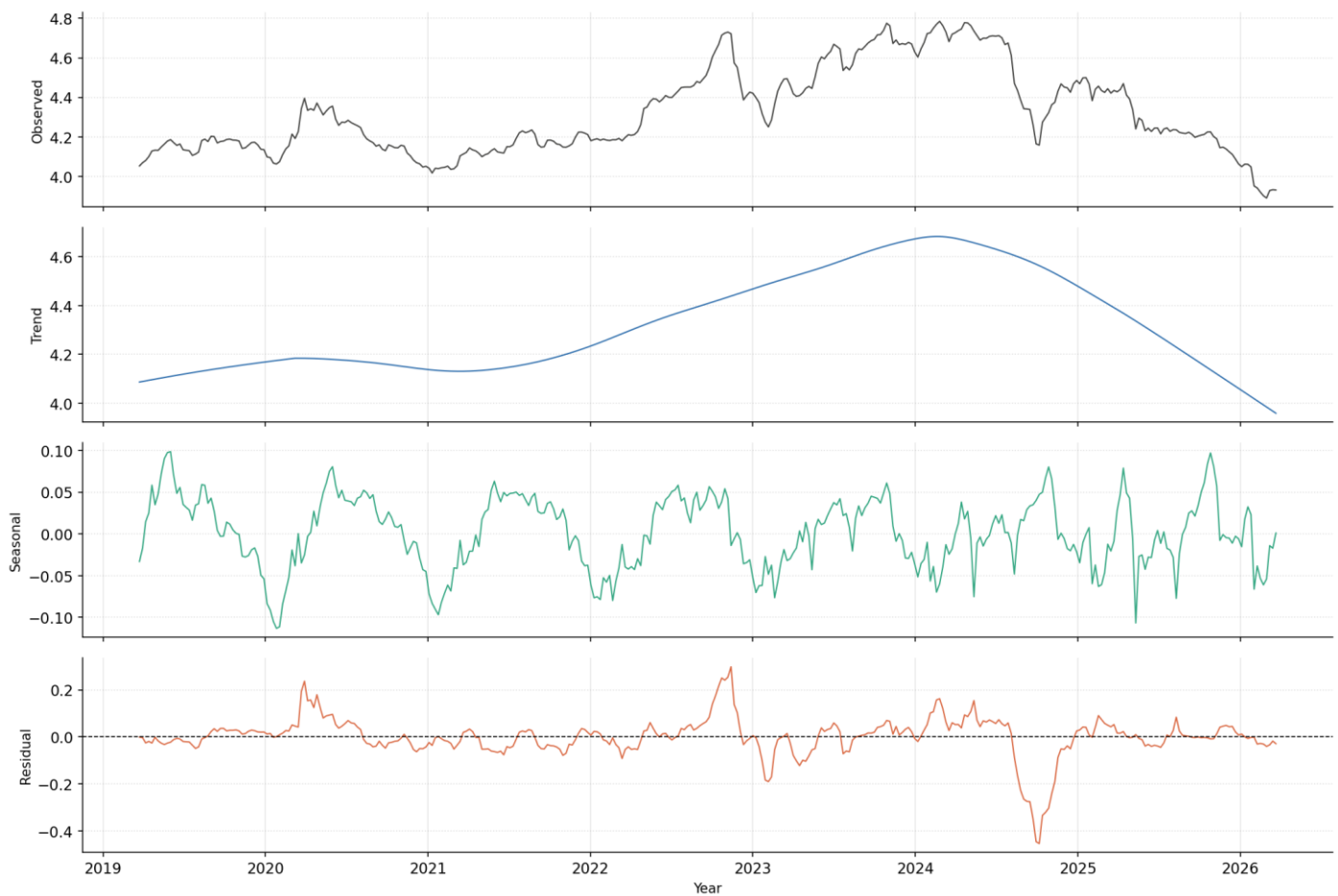


Figure 5b: STL Decomposition for MYR/USD Exchange Rate

In Conclusion, based on this study the STL is the best preferred decomposition method as the 7 year time series contained 2 significant structural shocks which are coinciding with the major global events such as the COVID-19 pandemic and geopolitical oil price war that could possibly distort results under the classical decomposition

2.4 Distribution and Correlation

The Brent crude oil prices are widely distributed between \$19 and \$128 per barrel with most values concentrated around \$60 to \$85 aligning with the mean and median. The distribution is slightly left-skewed with a secondary cluster of low prices below \$45 likely linked to the 2020 COVID-19 pandemic which reflecting the overall high volatility.

The MYR/USD exchange rate has shown a bimodal distribution with peaks around RM4.10 to RM4.20 and RM4.40 to RM4.75 indicating two exchange rate explanation meaning a stronger MYR during the pre-2022 and a weaker MYR at post-2022 due to US rate hikes. The range spans roughly RM3.88 per USD to RM4.80 per USD.

The **correlation ($r = 0.417$)** indicates a moderate positive relationship where higher Brent prices are associated with a weaker MYR. However, the relationship is not strongly linear suggesting other

macroeconomic factors such as US interest rates and other global conditions play a significant role where the strength of USD outweighed the oil price effects.

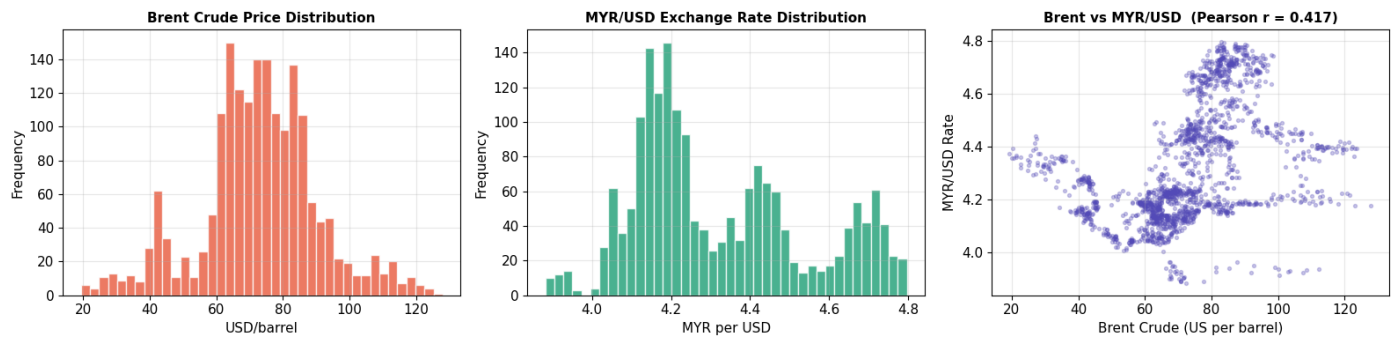


Figure 6: Distribution and Correlation for Brent vs MYR/USD

Step 3: Methodology

3.1 Stationary Tests

3.1.a Stationary Test using ADF and KPSS

In order to determine either which series require differencing before modelling 2 complementary stationary test are applied which are the Augmented Dicky-Fuller (ADF) test and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test.

ADF test

1. Null hypothesis: The series is nonstationary and has a unit root
2. Alternate Hypothesis: The series is stationary

KPSS Test

1. Null hypothesis: The series is stationary
2. Alternate Hypothesis: The series is non-stationary
- 3.

Table of **Original Series** for both Brent Crude Oil and MYR/USD Exchange Rate

Series	ADF Statistics	ADF P-Value	KPSS Statistics	KPSS P-value	Conclusion
Original Brent Crude Oil	-1.5663	0.5005	1.7582	0.0100	Both test converge, based on ADF P value we fail to reject null hypothesis and KPSS P value we reject null hypothesis. Conclude that, Non-Stationary
Original MYR/USD Exchange Rate	-1.4436	0.5613	1.9953	0.0100	Both test converge, based on ADF P value we fail to reject null hypothesis and KPSS we P value reject null hypothesis. Conclude that, Non-Stationary

Table of **After First Differencing** for both Brent Crude Oil and MYR/USD Exchange Rate

Series	ADF Statistics	ADF P-Value	KPSS Statistics	KPSS P-value	Conclusion
Brent Crude Oil After 1 st Differencing	-8.9634	0.0000	0.1134	0.1000	Both test converge, based on ADF P value we reject null hypothesis and KPSS P value we fail to reject null hypothesis. Conclude that, Stationary
MYR/USD Exchange Rate After 1 st Differencing	-19.0327	0.0000	0.3071	0.1000	Both test converge, based on ADF P value we reject null hypothesis and KPSS P value we fail to reject null hypothesis. Conclude that, Non-Stationary

3.1.b Visualize Original vs 1st Difference Series

Figure 7 compares the original and first-differenced series to illustrate the effect of differencing on stationarity.

For **Brent Crude Oil**, the original series is non-stationary which fluctuating between \$20 and \$128 per barrel with a prolonged trough in 2020 and a sharp peak in 2022. First differencing stabilizes the mean around zero with most daily changes within +-\$5 per barrel and occasional spikes with -\$17 per barrel happened in early 2022 and +\$8 in middle of 2022 reflecting major global events. The differenced series is stationary despite these volatility spikes. For **MYR/USD**, the original series shows a clear upward trend from RM4.05 per USD in 2019 to RM4.75 per USD in late 2023 then a sharp decline below RM3.90 per USD by early 2026 indicating non-stationarity. First differencing removes the trend and produces fluctuations around zero with most changes within +/-0.025 and occasional spikes with -0.10 in late 2023 and -0.09 in late 2022 to 2023. The differenced series confirms stationarity. These visual results align with ADF and KPSS tests confirming that $d = 1$ is sufficient for stationarity before ARIMA modelling.

Figure 7: Original vs First-Differenced Series

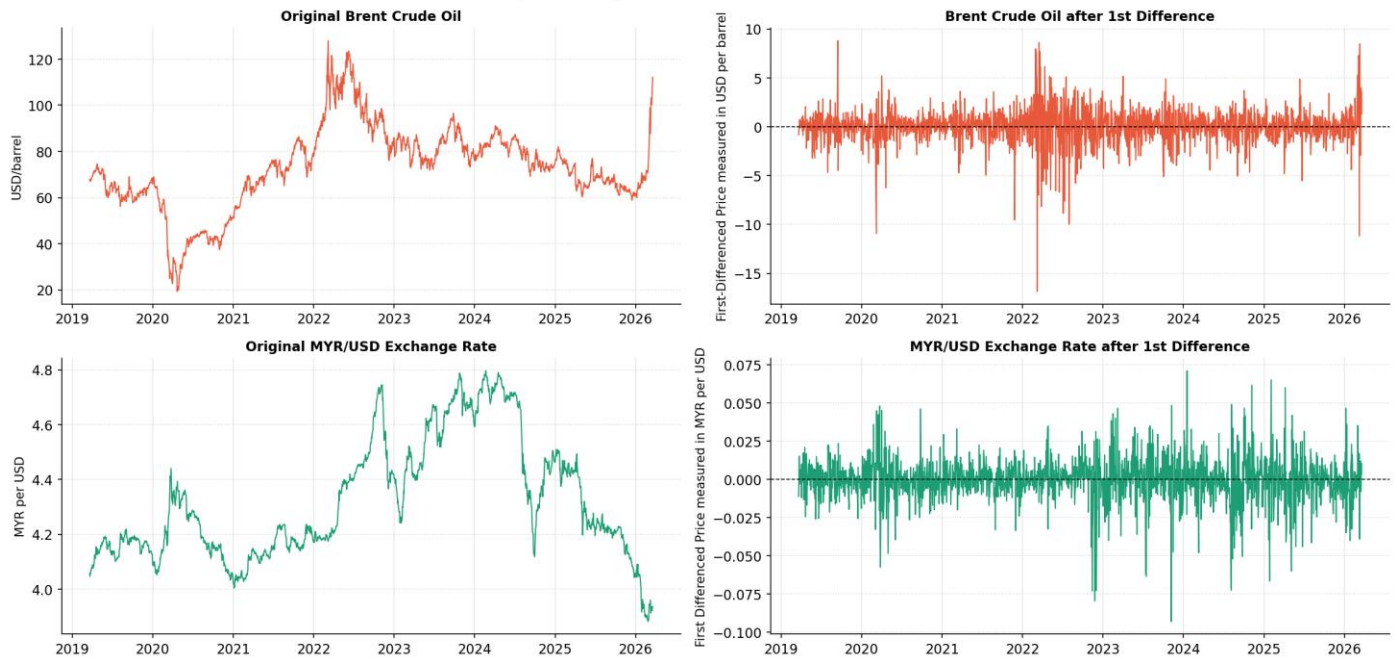


Figure 7: Original vs First-Differenced Series

3.2 Modelling and Model Selections

3.2.a ACF and PACF Analysis

The autocorrelation function (ACF) and partial autocorrelation function (PACF) plots are computed on the first differenced series to guide the selection of AR order(p) and MA order(q) for the ARIMA model. Both plots display 40 lags with 95% confidence intervals shaded in blue regions and if there are any spike exceeding the confidence band is consider as statistically significant.

Table of the Summary for Model Selection

Series	ACF Pattern	PACF Pattern	Order
Brent Crude Oil	Cut off after lag 0	Cuts off after lag 0	ARIMA($p \leq 2, 1, q \leq 2$)
MYR/USD Rate	Cuts off after lag 0	Cuts off after lag 0	ARIMA($p \leq 2, 1, q \leq 2$)

Figure of ACF & PACF Brent Crude Oil after 1st Difference

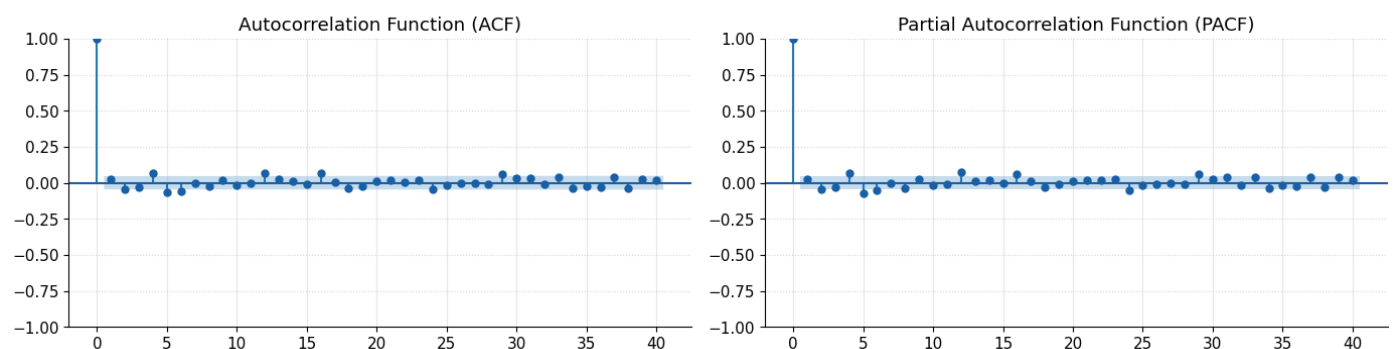


Figure 8a: ACF and PACF of Brent Crude Oil

Figure of ACF & PACF USD/MYR Exchange Rate after 1st Difference

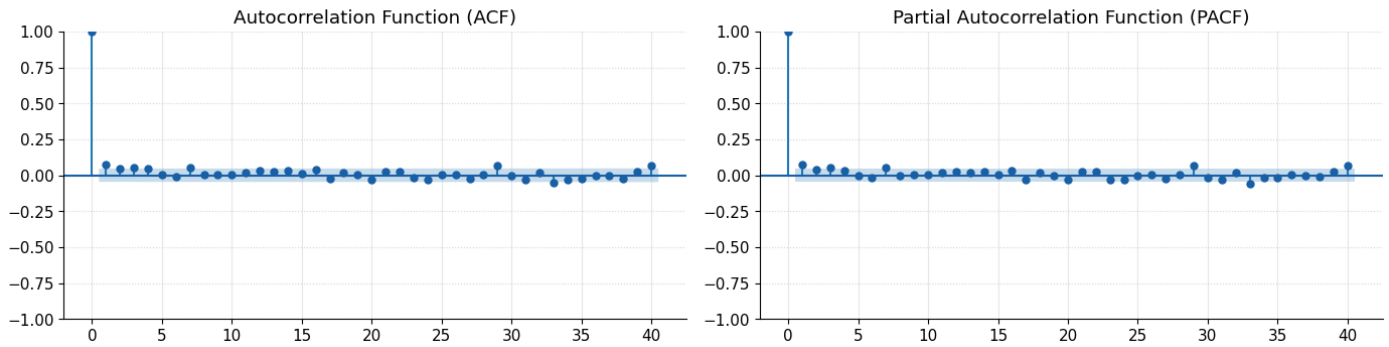


Figure 8b: ACF and PACF of MYR/USD Exchange Rate

3.2.b Auto ARIMA using AIC Grid Search

The `auto_arima` stepwise search with $d = 1$ identified the best ARIMA models for each series by minimizing AIC.

For the Brent Crude Oil time series the best mode is **ARIMA(0,1,0)** which is identified as a random walk with the AIC is 7297.12 and BIC is 7302.63. Random Walk simply meaning that tomorrow's price is best predicted by today's price and no AR or MA terms add explanatory power. Based on the SARIMAX result, the residual shows no autocorrelation with the Ljung-Box of 0.24 but are not normal given that the Jarque-Bera (JB) has the p value of 0.00 with heavy tail identify by the kurtosis which is 12.46 most likely due to the extreme event like COVID-19 and other geopolitical shocks.

For the **MYR/USD Exchange Rate** time series the best model is **ARIMA(4,1,0)** with no intercept given that the AIC is -10330.35 and BIC is -10302.80. Given the ARIMA(4,1,0) meaning that daily exchange rate changes are weakly influenced by the previous four days with the strongest effect from lag 1 with $AR1 = 0.0668$ and $p < 0.001$. Lag 4 is borderline insignificant. Based on the SARIMAX results the residuals are uncorrelated given that the Ljung-Box has p value of 1.00 but non-normal given that the Jarque-Bera has p value of 0.00 and heteroskedastic with p value of 0.00 reflecting occasional large movements. Residual variance is small with σ^2 of 0.0002.

Key Contrast are Brent behaves as a **pure random walk** with no short-term memory while the **MYR/USD Exchange rate** shows **mild short-term autocorrelation**

Summary table of `autoarima` for Brent Crude Oil and the MYR/USD Exchange Rate

Series	Best Auto Arima	AIC	BIC
Brent Crude Oil	ARIMA(0,1,0)	7,297.12	7,302.63
MYR/USD Exchange Rate	ARIMA(4,1,0)	-10,330.35	-10,302.80

3.2.c ARIMA Model Selection: Comparison of Box-Jenkins vs Auto ARIMA

Two approaches are applied to identify the optimal ARIMA order which is the Box-Jenkins method guided by the ACF and PACF with $p \leq 2$ and $q \leq 2$ and Auto ARIMA stepwise search with larger candidate space.

Summary table of difference approach with their selected model based on optimal AIC and BIC values.

Series	Approach	Selected Model	AIC	BIC
Brent Crude Oil	Box Jenkins with $p \leq 2$ and $q \leq 2$	ARIMA(0,1,0)	7297.12	7302.63
Brent Crude Oil	Auto ARIMA (max_p=5)	ARIMA(0,1,0)	7297.12	7302.63
MYR/USD Exchange Rate	Box Jenkins with $p \leq 2$ and $q \leq 2$	ARIMA(1,1,1)	-10333.23	-10316.70
MYR/USD Exchange Rate	Auto ARIMA (max_p=5)	ARIMA(4,1,0)	-10330.35	-10302.80

For **Brent Crude Oil** both approach converge and selected ARIMA(0,1,0) with identical AIC of 7291.12 and BIC of 7302.63 meaning that the oil price follows a pure random walk with no exploitable autocorrelation which is consistent with the ACF and PACF analysis carried out previously.

For **MYR/USD Exchange rate** the Box-Jenkins selected the ARIMA(1,1,1) with AIC of -10333.23 combining with one AR term indicate a short term momentum and one MA term which indicate a forecast error corrections. The auto ARIMA selected ARIMA(4,1,0) with AIC of -10330.35 given that it has a higher order pure AR therefore give slightly worse fit. Based on the results, the Box-Jenkins model for this time series are selected due to the lower AI.

Table of Final Selected Model for Forecasting

Series	Selected Model	Final Model	AIC	BIC
Brent Crude Oils	Auto ARIMA where both method agree	ARIMA(0,1,0)	7297.12	7302.63
MYR/USD Exchange Rate	Box-Jenkins	ARIMA(1,1,1)	-10333.23	-10316.70

3.2.d SARIMA Model Selection: Comparison of Box-Jenkins vs Auto SARIMA

The SARIMA model selection extended the ARIMA framework to include seasonal components such as (P,D,Q)[m] with weekly cycle of $m=5$. Non seasonal order (p,d,q) was inherited from the final ARIMA selections.

Table of SARIMA Non-seasonal constraints

Series	Selected ARIMA	SARIMA Non-Seasonal Constraints
Brent Crude Oil	ARIMA(0,1,0)	$p \leq 0, q \leq 0, D=1, P \leq 2, Q \leq 2, m=5$
MYR/USD Exchange Rate	ARIMA(1,1,1)	$p \leq 1, q \leq 1, D=1, P \leq 2, Q \leq 2, m=5$

Summary table of difference approach with their selected model based on optimal AIC and BIC values.

Series	Approach	Selected Model	AIC	BIC
Brent Crude Oil	Box-Jenkins with the Constrained	SARIMA(0,1,0)(2,1,0)[5]	7820.86	7837.38
Brent Crude Oil	Auto SARIMA	SARIMA(0,1,0)(2,1,0)[5]	7820.86	7837.38
MYR/USD Exchange Rate	Box-Jenkins with the Constrained	SARIMA(1,1,0)(2,1,0)[5]	-9778.52	-9756.49
MYR/USD Exchange Rate	Auto SARIMA	SARIMA(1,1,0)(2,1,0)[5]	-9778.52	-9756.49

For Brent Crude Oil, the final selected model is SARIMA(0,1,0)(2,1,0)[5] with the AIC of 7820.86 and BIC with 7837.38. Meaning that this series retained its random walk behaviour with non-seasonal AR and MA terms there are 2 seasonal AR terms at lag 5,10 and 15 capturing weekly mean reverting pattern.

For MYR/USD Exchange Rate, the selected model is SARIMA(1,1,0)(2,1,0)[5] with the AIC of -9778.52 and BIC of -9756.49. The non seasonal AR(1) term from ARIMA selection while 2 seasonal AR term capture the same weekly cycle. The MA(1) term from ARIMA is no longer needed once seasonality is included.

Both Box-Jenkins and Auto SARIMA agree on the final SARIMA models for both series therefore producing identical AIC and BIC values because both series share the same seasonal structure of (2,1,0)[5] indicate a 2 week seasonal AR pattern with weekly differencing.

Final Summary of SARIMA Selection

Series	Selected Method	Final Model	AIC	BIC
Brent Crude Oil	Both Agree	SARIMA(0,1,0)(2,1,0)[5]	7820.86	7837.38
MYR/USD Exchange Rate	Both Agree	SARIMA(1,1,0)(2,1,0)[5]	-9778.52	-9756.49

Step 4: Diagnostics

4.1 Model Comparison based on AIC and BIC

The AIC and BIC of the final selected ARIMA and SARIMA models are compared for each of the series. A lower AIC or BIC indicate a better trade-off between model fit. For both of the time series, the ARIMA outperforms SARIMA on both AIC and BIC. Therefore both series selected ARIMA.

Table of Comparison of SARIMA and ARIMA model

Series	Model	AIC	BIC
Brent Crude Oil	ARIMA(0,1,0)	7297.12	7302.63
Brent Crude Oil	SARIMA(0,1,0)(2,1,0)[5]	7820.86	7837.38
MYR/USD Exchange Rate	ARIMA(1,1,1)	-10316.70	-10316.70
MYR/USD Exchange Rate	SARIMA(1,1,0)(2,1,0)[5]	-9778.52	-9756.49

4.2 Interpretation of Summaries of Final Selected Model

Based on the AIC and BIC result the ARIMA(0,1,0) is selected for Brent Crude Oil, this model is a pure random walk with only residual variance of σ^2 is approximately 3.18 implying daily shock of $\pm \$1.78$ per barrel. For this model, it indicates that tomorrow price is best predicted as today price plus a random shock.

Table of summary result of ARIMA(0,1,0) for Brent Crude Oil

Parameter	Coefficient	P value	Interpretation
Sigma²	3.1812	0.000	Residual variance given that σ^2 is approximate 3.18 which is around \$1.78 per barrel daily stock

For MYR/USD exchange rate, the best model based on AIC and BIC is the ARIMA model which is ARIMA(1,1,1). The model contained 3 estimated parameters.

Table of summary results of ARIMA(1,1,1)

Parameter	Coefficient	P value	Interpretation
ar.L1	0.7556	0.000	Strong positive AR(1) meaning today's change is significantly influenced by yesterday's change
ma.L1	-0.6870	0.000	The negative MA(1) coefficient indicates that forecast errors from the previous day are partially reversed in the current period. In practical terms, if the model overestimates yesterday it will adjust downward today and vice versa helping to stabilise short-term fluctuations.
sigma²	0.0002	0.000	Very small residual variance as the sigma is approximate 0.014 MYR per USD daily

4.3 Residual Diagnostics

In this subsection, I will include the summary of the diagnostics and their corresponding diagnostic plots. Although SARIMA model was not selected based on performance metrics since their AIC value were slightly higher than those of the ARIMA models we are still interested in evaluating how SARIMA performs in forecasting for study purposes.

Final ARIMA and SARIRMA Model comparison for Brent Crude Oil and MYR/USD

Series	Model	Is ACF Clean	Normality	Selected
Brent Crude Oil	ARIMA(0,1,0)	Clean	Non-normal with heavy tails	Yes
Brent Crude Oil	SARIMA(0,1,0)(2,1,0)[5]	Spikes at lag 5, 10 and 15	Non-normal with heavy tails	No
MYR/USD Exchange Rate	ARIMA(1,1,1)	Clean	Near normal	Yes
MYR/USD Exchange Rate	SARIMA(1,1,0)(2,1,0)[5]	Spike at lag 5	Near normal	No

Figure: Residual Diagnostics of ARIMA Brent Crude Oil (0, 1, 0)

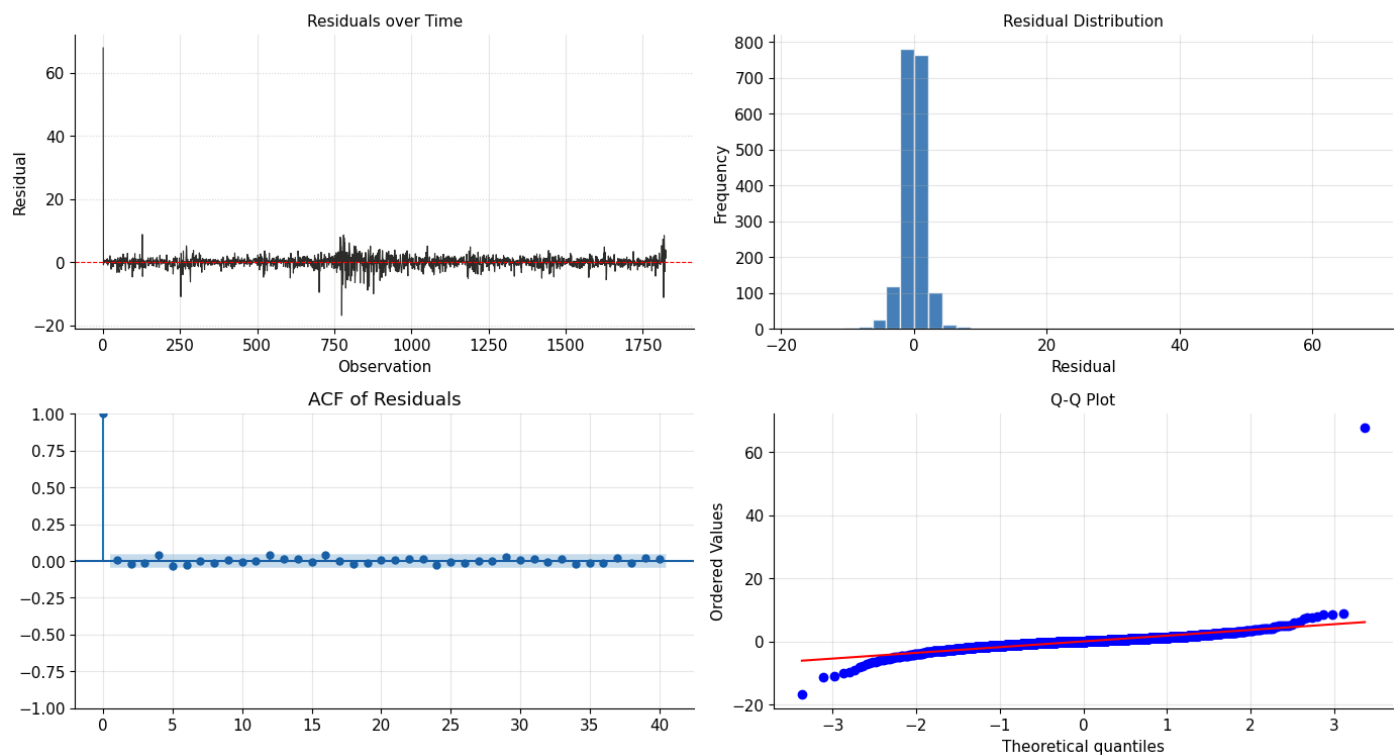


Figure 9a: Residual Diagnostics for ARIMA(0,1,0) Brent Crude Oil

Figure: Residual Diagnostics of SARIMA Brent Crude Oil (0, 1, 0)x(2, 1, 0, 5)

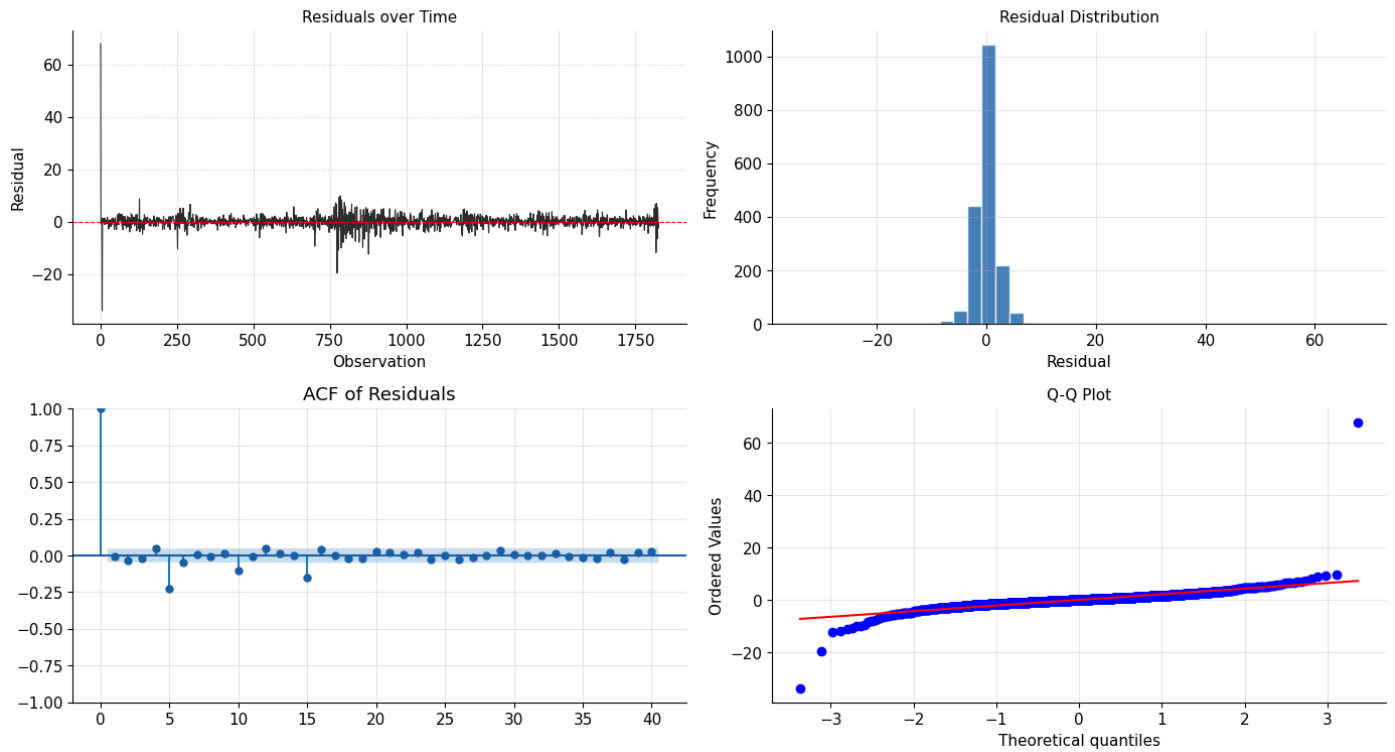


Figure 9b: Residual Diagnostics for SARIMA(0,1,0)(2,1,0)[5] Brent Crude Oil

Figure: Residual Diagnostics of ARIMA MYR/USD Rate (1, 1, 1)

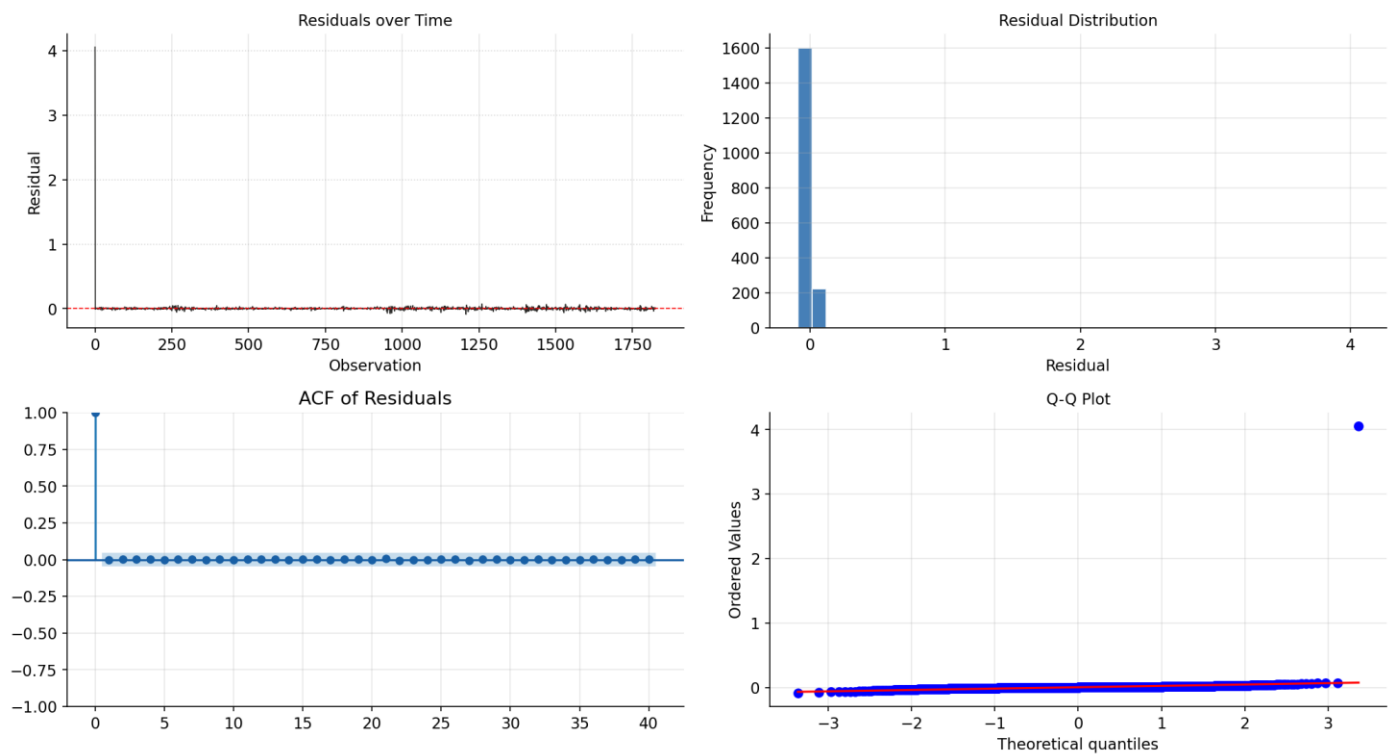


Figure 9c: Residual Diagnostics for ARIMA(1,1,1) MYR/USD Rate

Figure: Residual Diagnostics of SARIMA MYR/USD Rate (1, 1, 0)x(2, 1, 0, 5)

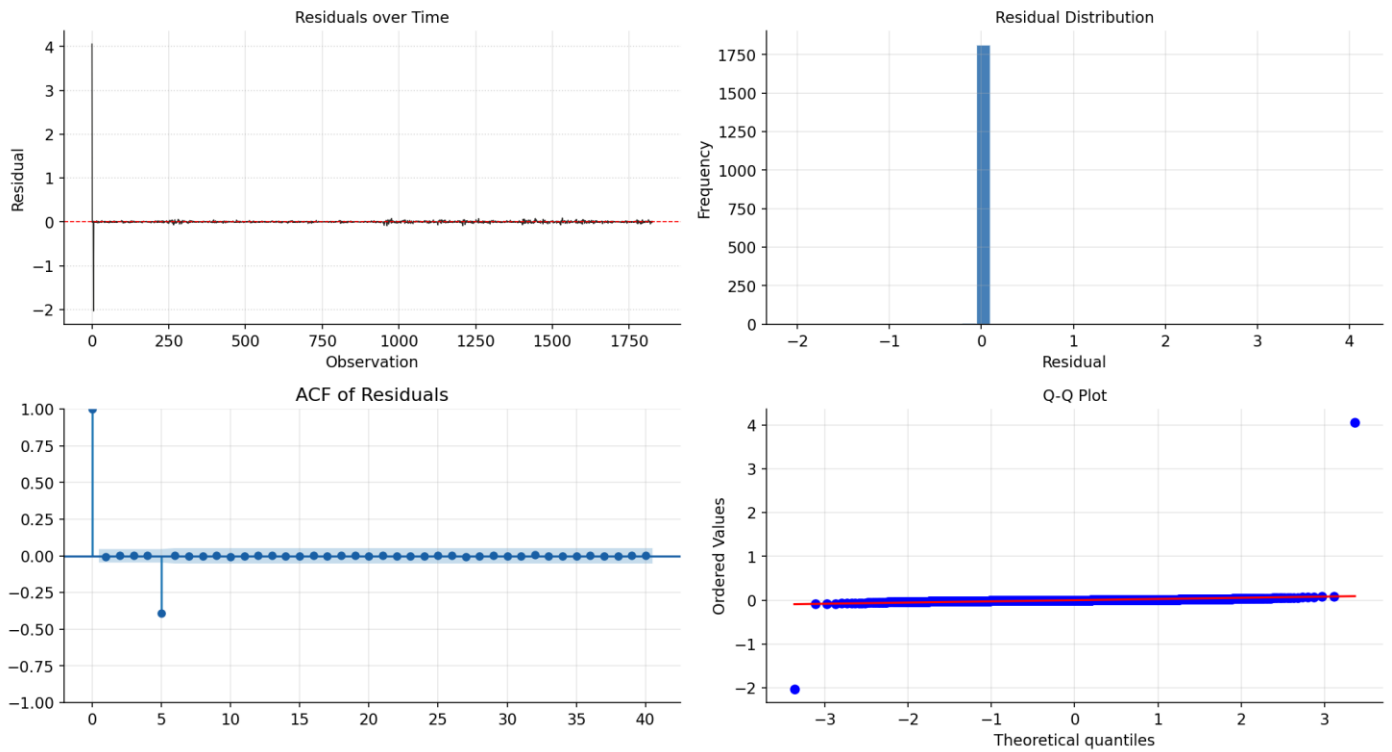


Figure 9d: Residual Diagnostics for SARIMA(1,1,0)(2,1,0)[5] MYR/USD Rate

Step 5: Forecasting

In this section, I developed 3 models and evaluated them over the same 30 day test period from 9th February 2026 to 20th March 2026. All the models are trained on 1797 observation from 21st March 2019 to 6th of February 2026 and the final 30 business days are reserved as test data.

5.1 1st Model ARIMA Forecast

ARIMA model selected from previous section are trained on all available observation except the final 30 business days which are reserved as test data. Unlike the conventional 80/20 training and test splits data, this approach maximizes the training information for the 30 days forecast

Table of Data Split and Model Specification for ARIMA forecast

	Brent Crude Oil	MYR/USD Exchange Rate
Model	ARIMA(0,1,0)	ARIMA(1,1,1)
Train	1797 observation from 21 st March 2019 to 06 th February 2026	1797 observation from 21 st March 2019 to 06 th February 2026
Test	30 observation from 09 th February 2026 to 20 th March 2026	30 observation from 09 th February 2026 to 20 th March 2026

The Brent Crude Oil with ARIMA (0,1,0) model produce a flat forecast at approximately \$68 per barrel across the 30 day period. This is consistent with the assumption of random walk where the best prediction of the future is simply the most recent observed price. The 96% confidence interval widens over time reflecting the increased forecast uncertainty given that at Day 1 the confidence interval is around range of \$60 to \$77 until Day 30 the range of 95% confidence interval increase to around range of \$50 to \$90.

By comparing the forecast and the actual performance for Brent Crude Oil with ARIMA(0,1,0). I noticed that during the test period from 9th February 2026 go 20th March 2026 the actual price diverged sharply with the forecast price rising from around \$68 to \$112 per barrel with an increase of around 65% within 30 days. This sharp increase is directly caused by geopolitical war shock happened in February 2026 which trigger a rapid increase in oil price. As a result, the actual price moves above the upper bound of 95% confidence interval by the middle of March. This structural break represents an extreme and exogenous event that lies outside the historical pattern captured by the model.

The MAPE of 15.36% confirm that the model significantly underestimates the magnitude of the price increase during this shock period

Table of forecast accuracy metrics for the Brent Crude Oil with ARIMA(0,1,0)

Metric	Value
MAE	14.7547
RMSE	20.8789
MAPE	15.36%

Figure: ARIMA(0, 1, 0) 30 Day Forecas Brent Crude Oil
Forecast Period: 09 Feb 2026 to 20 Mar 2026



Figure 10a: ARIMA(0,1,0) Forecast for Brent Crude Oil

For the MYR/USD exchange rate with ARIMA(1,1,1), it produces a near flat forecast at around 3.95 MYR per USD with the AR(1) coefficient of 0.8437 and MA(1) coefficient of -0.7793 which largely cancelling each other out therefore produce minimal drift beyond the first step. The 95% confidence interval are at the range of around 3.75 MYR per USD to 4.15 MYR per USD.

The actual MYR/USD exchange rate during the test period is fluctuates between 3.88 MYR per USD and 3.96 MYR per USD. The remaining are largely within the 95% confidence interval throughout the 30-day period.

A MAPE of 0.72% showed that the model tracks the MYR/USD rate with high relative accuracy over the 30 day period.

Table of forecast accuracy metrics for the MYR/USD ARIMA(1,1,1)

Metric	Value
MAE	14.7547
RMSE	20.8789
MAPE	15.36%

Figure: ARIMA(1, 1, 1) 30 Day Forecas MYR/USD Exchange Rate
Forecast Period: 09 Feb 2026 to 20 Mar 2026



Summary table of ARIMA forecast

Series	Model	MAE	RMSE	MAPE
Brent Crude Oil	ARIMA(0,1,0)	14.7547	20.8789	15.36%
MYR/USD Exchange Rate	ARIMA(1,1,1)	0.0280	0.0338	0.72%

In conclusion, for this ARIMA forecast both series is extremely different. The ARIMA(1,1,1) perform exceptionally well for MYR/USD given that the test period is characterized by a relatively stable exchange rate that remain within the model predicted uncertainty 95% confidence intervals region. In contrast, the ATIMA(0,1,0) performs poorly for Brent Crude Oil due to a geopolitical shock during the test period this represent a structural break that lies beyond the predictive capability of the model trained solely on historical data.

5.2 2nd Model SARIMA Forecast

SARIMA forecast is included to compare the outcome of the forecast to other forecasting method applied in this project.

Table of Data Split and Model Specification for SARIMA forecast

	Brent Crude Oil	MYR/USD Exchange Rate
Model	SARIMA(0,1,0)(2,1,0)[5]	SARIMA(0,1,0)(2,1,0)[5]
Train	1797 observation from 21 st March 2019 to 06 th February 2026	1797 observation from 21 st March 2019 to 06 th February 2026
Test	30 observation from 09 th February 2026 to 20 th March 2026	30 observation from 09 th February 2026 to 20 th March 2026

For Brent Crude oil with SARIMA(0,1,0)(2,1,0)[5] the forecast exhibits a mild upward drift rising from around \$68 per barrel to around \$73 by 20th March 2026. The 95% confidence interval widens substantially over the forecast period given that at day 1 of the forecast at around range of \$58 to \$80 to day 30 with around \$40 to \$145. This reflects increasing uncertainty over time. In contrast, the actual price surged from around \$68 to around \$122 per barrel exceeding the upper bound of the confidence interval by middle of March this suggests the presence of an unexpected geopolitical shock not captured by the model.

Table of forecast accuracy metrics for the **Brent Crude oil with SARIMA(0,1,0)(2,1,0)[5]**

Metric	Value
MAE	12.7846
RMSE	18.5836
MAPE	13.19%

Figure: SARIMA(0, 1, 0)x(2, 1, 0, 5) 30 Day Forecast Brent Crude Oil
Forecast Period: 09 Feb 2026 to 20 Mar 2026

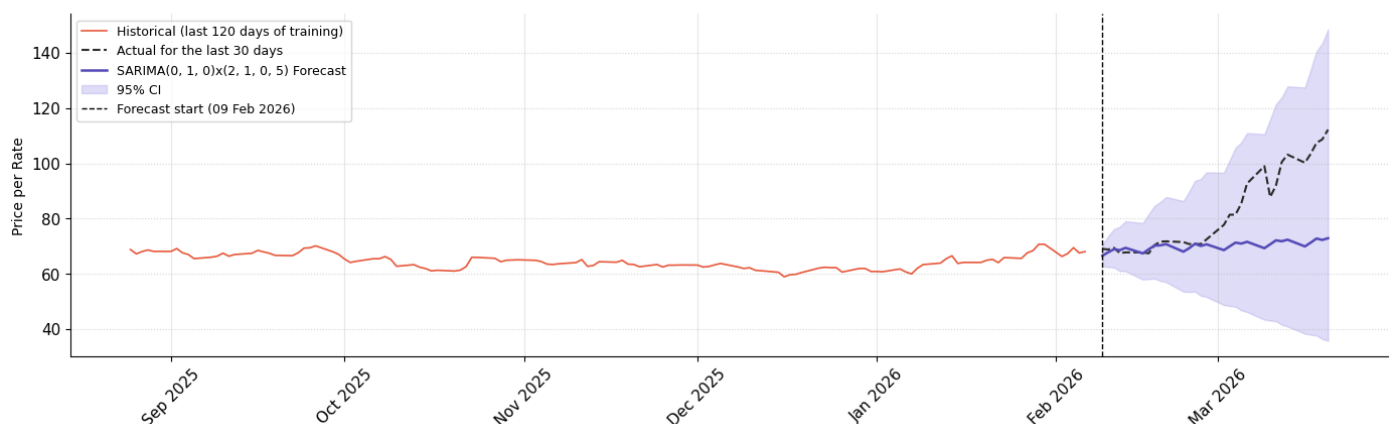


Figure 11a: SARIMA Forecast for Brent Crude Oil

For MYR/USD Exchange Rate with SARIMA(0,1,0)(2,1,0)[5] the forecast starts at around 3.93 MYR per USD and trends downward to about 3.77 by 20 Mar 2026. The 95% confidence interval expands as from Day 1 at the range of around 3.85 MYR per USD to 4.00 MYR per USD up until day 30 with range of around 3.40 MYR per USD to 4.10 MYR per USD. The actual exchange rates fluctuate between around

3.88 and 3.96 MYR/USD, remaining largely **within the 95% confidence interval** during the early forecast period before slightly breaching the lower bound later in February.

Table of forecast accuracy metrics for the **Brent Crude oil with SARIMA(0,1,0)(2,1,0)[5]**

Metric	Value
MAE	0.0741
RMSE	0.0954
MAPE	1.89%

Figure: SARIMA(1, 1, 0)x(2, 1, 0, 5) 30 Day Forecast MYR/USD Exchange Rate
Forecast Period: 09 Feb 2026 to 20 Mar 2026



Figure 11b: SARIMA Forecast for USD/MYR Exchange Rate

Summary table of ARIMA and SARIMA forecast

Series	ARIMA RMSE	SARIMA RMSE	Better Model
Brent Crude Oil	20.8789	18.5836	SARIMA
MYR/USD Exchange Rate	0.0338	0.0954	ARIMA

5.3 3rd Model FNN Forecast (Keras + Dropout + Early Stopping)

A Feedforward Neural Network (FNN) is applied as the third forecasting model using a 20 day lag window to create supervised input and output pair from the log-transformed series. The same train-test split is maintained for consistency of 1777 training samples with the final 30 business day from 9th February 2026 to 20th March is reserved as test data.

The model architecture is Input(20) to Dense(64, ReLU) + L2 + Dropout(0.2) to Dense(32, ReLU) + L2 + Dropout(0.2) to Dense(1).

L2 regularisation and Dropout rate=0.2 are applied to both hidden layers to reduce overfitting. Early stopping monitors validation loss with patience=20 restoring the best weights when validation loss stops improving.

The FNN forecast of Brent Crude oil has the training loss curve located at the left panel show rapid convergence with the first 10 epoch with both train and validation loss stabilizing near zero. Early stopping is triggered at epoch 68 indicating the model converged effectively without overfitting

The forecast plot at right panel shows the FNN successfully captures the sharp increase in price momentum during the test period with corresponding actual price increase from approximately \$68 to \$112 per barrel closely. This contrasts with the flat forecasts produced by ARIMA and SARIMA demonstrating the FNN ability to detect and extrapolate strong directional trends encoded in the 20 day lag window

Table of forecast accuracy metrics for the **FNN Brent Crude oil**

Metric	Value
MAE	4.9009
RMSE	6.3660
MAPE	5.58%

Figure: FNN 30 Day Forecast Brent Crude Oil
Forecast Period: 09 Feb 2026 to 20 Mar 2026

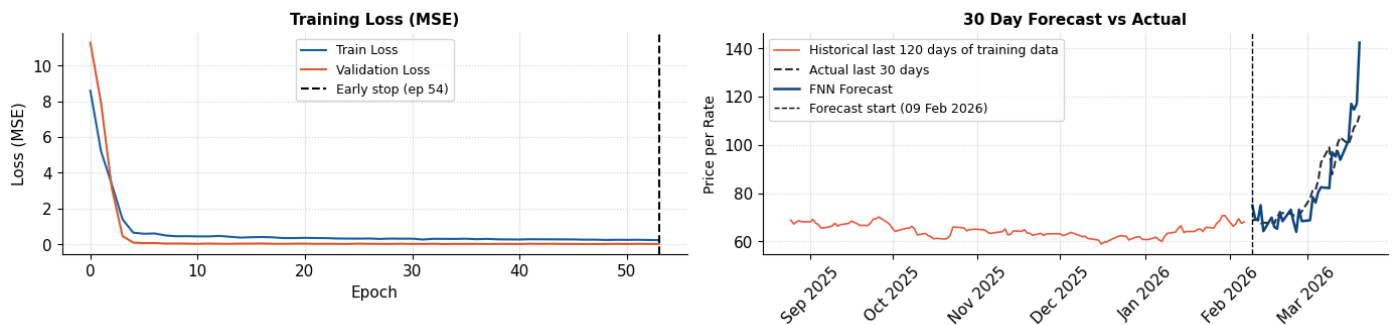


Figure 12a: FNN Forecast for Brent Crude Oil

For the FNN of MYR/USD Exchange rate their training loss curve shows rapid convergence within the first 20 epochs with early stopping triggered at **epoch 304** indicating a more complex optimisation compared to Brent. Both train and validation loss converge closely confirming no significant overfitting.

The forecast plot shows the FNN producing a relatively stable forecast around 3.97 MYR per USD to 4.00 MYR per USD which is close to the actual exchange rate which fluctuated between approximately 3.88 MYR per USD and 3.96 MYR per USD during the test period.

Table of forecast accuracy metrics for the **FNN MYR/USD Exchange rate**

Metric	Value
MAE	0.0332
RMSE	0.0377
MAPE	0.85%

Figure: FNN 30 Day Forecast MYR/USD Exchange Rate
Forecast Period: 09 Feb 2026 to 20 Mar 2026

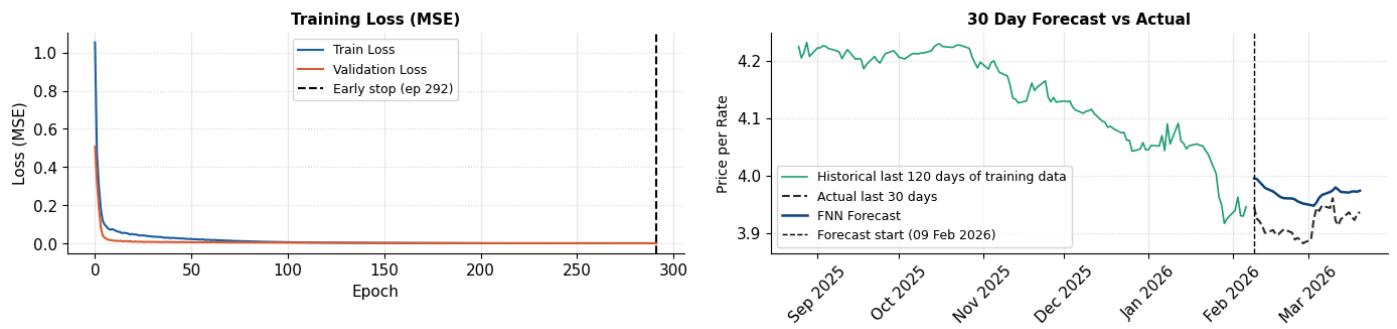
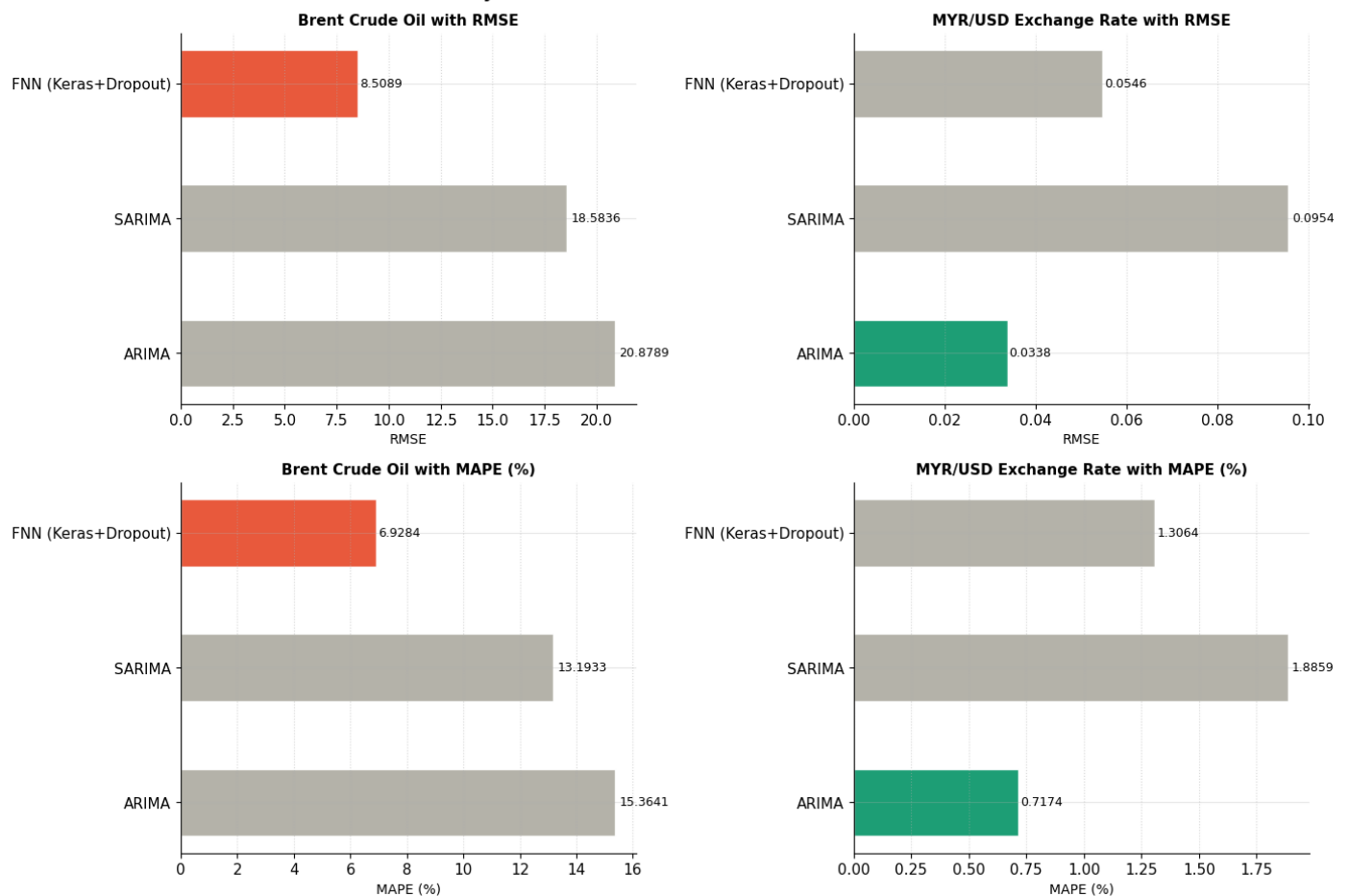


Figure 12b: FNN Forecast for USD/MYR Exchange Rate

5.4 Summary of Model Comparison and Forecast Accuracy

The forecast accuracy of all three models is evaluated using MAE, RMSE and MAPE on the held-out test set of 30 business days ending 20 Mar 2026. The highlighted bar in each chart indicates the best performing model for that metric.

Figure: Model Comparison for ARIMA vs SARIMA vs FNN
30 Day Forecast from 09 Feb 2026 to 20 Mar 2026



For Brent Crude Oil, FNN is the best performing model across all other model. With an RMSE of 6.37 and MAPE of 5.58%, the FNN outperforms SARIMA around **65% on RMSE** and ARIMA by approximately **70% on RMSE**. The FNN's ability to capture the non-linear upward price momentum through its 20 day lag window explains this superior performance during the geopolitical price shock of Feb 2026 to Mar 2026.

Table of model performance for Brent Crude Oil

Model	MAE	RMSE	MAPE(%)
ARIMA(0,1,0)	14.7547	20.8789	15.36%
SARIMA(0,1,0)(2,1,0)[5]	12.7846	18.5836	13.19%
FNN (Keras + Dropout + Early Stopping)	4.9009	6.3660	5.58%

For MYR/USD Exchange Rate the ARIMA(1,1,1) model is the best model across all other model by achieving a MAPE of just 0.72% tracks the actual exchange rate with high relative accuracy during the stable test period. The FNN performs closely behind ARIMA with a MAPE of 0.85% confirming that the lag-based neural network also captures the exchange rate behaviour well. SARIMA performs worst with a MAPE of 1.89% because the seasonal differencing introduces an unnecessary downward drift that diverges from the relatively stable actual values.

Table of Overall Final Selected Forecasting Model

Series	Best Model	RMSE	MAPE
Brent Crude Oil	FNN (Keras + Dropout + Early Stopping)	6.3660	5.58%
MYR/USD Exchange Rate	ARIMA(0,1,0)	0.0338	0.72%

The results reveal an important finding **no single model dominates across both series**. This highlights an important idea in time series forecasting where choosing **the right model depends more on the behaviour of the data and the forecasting period than on how complex the model is**.

Step 6: Conclusion

There is **limitation** presence for this time series analysis project, both the COVID-19 crash occurred at 2020 and other geopolitical war as well as the recent 2026 geopolitical shock represent structural breaks that ARIMA and SARIMA models cannot anticipate. These models assume a stable data generating process making them inherently unsuitable for forecasting during black swan events. Unlike ARIMA and SARIMA, **the FNN does not produce confidence intervals** making uncertainty quantification impossible without additional techniques such as Monte Carlo Dropout or bootstrap resampling. This limits the practical usefulness of the FNN forecast for risk management applications.

Both ARIMA and FNN models rely solely on the historical price series. In reality, Brent Crude Oil is influenced by OPEC production decisions, US dollar strength, geopolitical risk indices and global economic indicators. Similarly, MYR/USD is driven by interest rate differentials, capital flows and trade balances. Incorporating these external variables could improve forecast accuracy.

The suggested future improvement for this projects are for Brent Crude Oil would be an ARIMA-GARCH combination given that ARIMA models the conditional mean while GARCH(1,1) models the time-varying conditional variance. This directly addresses the heteroskedasticity limitation and would produce confidence intervals that widen appropriately during high-volatility periods such as the 2020 and 2022 shocks. Incorporating external regressors such as the WTI-Brent spread, US Federal Reserve interest rate decisions, OPEC production announcements and geopolitical risk indices into an ARIMAX framework could improve the time series forecast model's ability to anticipate structural shifts.

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